Emerging technologies and strategies in epilepsy surgery: toward personalised medicine

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Epilepsy surgery, the treatment of choice for drug-resistant focal epilepsy, is evolving rapidly. This progress is driven by a growing interest in the network theory of epilepsy, advances in data-driven models, and a focus on personalised treatment approaches. As a result, treatment options have expanded to include minimally invasive procedures, neurostimulation devices, and network-based interventions. Predicting surgical outcomes—such as seizure freedom and neuropsychological effects—remains challenging but is improving through advances in computational technology and molecular research, paving the way for more precise surgery. Despite these advancements, disparities in access to treatments persist, particularly in resource-scarce settings, highlighting the need for systemic solutions to improve access. Emerging research into genetic and multi-omic markers might assist in tailoring treatments and improving the prediction of outcomes. Future directions include integrating minimally invasive techniques, refining neuromodulation strategies, and leveraging molecular and computational tools to optimise patient care. Multidisciplinary collaboration will be essential to overcome challenges, reduce disparities, and advance surgical outcomes for patients with drug-resistant epilepsy worldwide.

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Introduction

Resective epilepsy surgery remains the gold standard and potentially curative treatment for a third of people with epilepsy who have drug-resistant focal seizures. The field is evolving with new surgical options, a better understanding of outcomes, and a focus on personalised care. Recognising epilepsy as a network disorder, coupled with concerns over cognitive and surgical complications, has spurred the development of minimally invasive alternatives and neurostimulation approaches.[1–3] Emerging evidence linking surgical outcomes with genetic and molecular mechanisms is advancing the potential for personalised surgery. Although the therapeutic landscape is expanding, the scarcity of robust comparative data complicates clinical decision making, leading to preference-driven rather than evidence-based care, with pronounced variability across centres.[4] In parallel, increasing disparities in global access to surgical care present substantial challenges.[5]

In this Review, we explore evolving practices in epilepsy surgery, address advancements in surgical technologies and outcomes, discuss disparities in care, and provide a vision for future innovations aimed at precision surgical planning through network and multi-omic approaches. We address three core challenge areas that mirror surgical care delivery: treatment options and decision making; predicting outcomes; and disparities in access and use.

Treatment options and decision making

Resection

Overview of surgical technique

Successful resection requires a well localised, safely accessible epileptogenic zone (panel 1). Patients with multifocal or eloquent focal epilepsies are typically ineligible. In adults with mesial temporal lobe epilepsy, hippocampal sclerosis is the most common pathology,[12] often treated with anterior temporal lobectomy or selective amygdalohippocampectomy, both requiring craniotomy. Anterior temporal lobectomy removes the amygdala, anterior hippocampus, and parts of the temporal neocortex,[13] with the resection extent guided by laterality, language dominance, and surgeon preference. Resection is done under general anaesthesia or awake language mapping, with or without intraoperative electrocorticography. Selective amygdalohippocampectomy restricts resection to the mesial temporal structures through a transcortical, subtemporal, or trans-Sylvian approach.[13]

Although larger resections can improve seizure control, they pose increased neurocognitive risk,[14] emphasising

Panel 1: The epileptogenic zone and innovations in how to define it

Definition

- First introduced by Talairach and Bancaud in the 1960s with stereo-EEG.[6]
- Defined as "the site of beginning of epileptic seizures" encompassing the seizure onset zone, and "zone of primary organization" where seizures propagate to produce clinical symptoms.[6]

Theoretical implications

- Removing the seizure onset zone is necessary, but not sufficient for seizure freedom.
- Seizure freedom demands a consistent, reproducible, and accurate removal of the hypothesised epileptogenic zone, including the difficult-to-define zone of primary organisation.

Practical implications

- Epilepsy surgery involves removal (resection) or ablation (laser interstitial thermal therapy) of this epileptogenic zone, or modulation of its underpinning electrical drivers (neuromodulation).
- Regardless of the approach, surgery aims to balance seizure freedom with minimal complications.

Measurement approaches

- Novel tools such as quantitative brain MRI[7] (eg, to detect subtle cortical volume loss outside the surgical site) and intracranial EEG biomarkers[8,9] (eg, removal of high frequency oscillations through temporal lobe surgery) have improved identification of the hypothesised epileptogenic zone and enhanced outcomes—especially when leveraged through deep learning on high quality open-source datasets.[10,11]

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the need to minimise the surgical footprint. Visualisation of cortical structures during resection enables safety and efficacy—eg, piriform cortex resection might enhance seizure freedom.[15] Further techniques, such as magnetic resonance tractography, might allow visualisation of visual and language pathways; however, implementation in a temporal lobectomy is limited by brain shift occurring during craniotomy.[16] Intraoperative electrocorticography, often used in anterior temporal lobectomy, helps to optimise the extent and functional safety.[17]

Extratemporal lobe epilepsy resections commonly target the frontal lobe, and less frequently, target the parietal lobe, occipital lobe, or insula,[18] and are customised as the extent of the epileptogenic zone varies. Multilobar resections or hemispherectomy can be considered with extensive epileptogenic cortex.

Temporal lobe epilepsy and extratemporal lobe epilepsy can have lesional or non-lesional MRI findings, with lesionectomy often sufficient in lesional cases unless a broader epileptogenic zone is suspected. Non-lesional cases typically need intracranial EEG to map the epileptogenic zone before resection.

Seizure outcomes

Decades of practice with resection have provided a comprehensive understanding of long-term outcomes. For refractory temporal lobe epilepsy, a randomised controlled trial in adult patients in Canada reported a number needed to treat of two,[19] favouring surgery over medical therapy. Seizure outcomes can be reported as complete seizure freedom after surgery; a multicentre, longitudinal cohort study from 18 European countries found that 70–80% of patients (adults and children) were seizure-free at 5 years post surgery,[20] although this drops to around 50% by 10 years.[21,22] If initial surgery is unsuccessful, reoperation remains an option. Success rates decline with each reoperation, with sustained seizure freedom reached in 30–50% of cases at 1 year postoperatively, declining to less than 40% by 5 years.[23] Seizure outcomes can also be expressed as a response-rate—defined as a 50% or higher reduction in seizure frequency compared with baseline. This measure is gaining popularity to facilitate objective outcome comparison with neuromodulation. In the largest retrospective cohort study to date, 90·4% (88·9–92·8) of patients (adults and children) post resection with ongoing seizures had 50% or greater reduction in seizure frequency over 9 years of follow-up. Less than 3% of patients worsened, suggesting clinically significant benefit even without full seizure freedom.[24]

A 2024 meta-analysis, which included observational studies (both prospective and retrospective), including more than 3000 patients, showed no significant difference between selective amygdalohippocampectomy and anterior temporal lobectomy in achieving seizure freedom, particularly for patients with mesial temporal lobe epilepsy with hippocampal sclerosis. However, selective amygdalohippocampectomy exhibited significantly lower risk of visual field deficits (risk ratio 0·87, 95% CI 0·78–0·97) than anterior temporal lobectomy.[25]

Extratemporal lobe epilepsy poses greater challenges than temporal lobe epilepsy due to complex connectivity and proximity to eloquent regions. Seizure freedom in frontal lobe resections ranges from 20% to 70% at 10 years, whereas posterior quadrant resections achieve approximately 40% at the same interval.[12]

Favourable outcomes are predicted by focal preoperative MRI features, such as hippocampal sclerosis, low-grade tumours, focal cortical dysplasia (particularly histopathologically type 2B), concordance between imaging and EEG, concordant PET hypometabolism,[26] complete resection, no intracranial evaluation, and non-dominant hemisphere resections.[27,28] Conversely, widespread, poorly defined, and localised abnormalities on MRI; focal to bilateral tonic–clonic seizures; and non-specific pathology are adverse prognostic factors.

Neuropsychological outcomes

Changes in neuropsychological function following resective surgery can be conceptualised as contraindications, complications, and costs of the procedure.[29] Contraindications refer to unacceptably high risks of cognitive impairment associated with the proposed surgery. Commonly, these are risks of profound memory impairment following anterior temporal lobectomy or potential effects on language function identified in the presurgical evaluation. When a presurgical investigation indicates high risks, surgery is contraindicated. Complications refer to unexpected cognitive impairments from infection, bleeds, or other surgical complications. Although these rates are low,[29] up to 60% of patients undergoing resection will have some postoperative decline in function in one or more cognitive domains.[30] This decline depends on the measurement used and the timescale for assessing outcomes. When expected, these deficits can be conceptualised as the neuropsychological costs of the procedure. Each prospective surgical candidate brings their own combination of risk and protective factors with respect to postoperative cognitive decline. Several models have been developed to predict postoperative outcomes[31,32] based on a patient's individual characteristics; however, a patient's baseline cognitive performance remains a key predictor of their postoperative state.[33] These predictions can be used to plan prehabilitation—the use of functions before they are lost—and to develop strategies and routines to prepare patients for any anticipated postoperative declines.[34]

Laser interstitial thermal therapy

Overview of surgical technique

Laser interstitial thermal therapy (LITT) is a stereotactic technique that delivers thermal ablation to a presumed epileptogenic zone. Preprocedure high-resolution MRI,

including gadolinium-enhanced T1-weighted and fluid attenuated inversion recovery sequences, guide stereotactic trajectory planning to avoid critical structures (blood vessels and ventricles), while optimising target cannulation.[35]

Based on the stereotactic system—traditional frames,[36] skull-mounted systems,[37] or robotic platforms[38]—the initial steps of LITT include general anaesthesia induction, marking entry site, shaving or parting hair, and drilling a craniotomy under stereotactic guidance. A skull-mounted bolt or platform guides the laser along the planned trajectory before holding the laser in place. The procedure is performed either in an operating room with subsequent transfer to MRI or entirely within an interventional MRI suite. Although a fully volumetric 3D rendering is not currently possible, two or three orthogonal planes (depending on the platform) can be monitored during the ablation. This quantitative feedback from magnetic resonance thermometry can provide surgeons with a reliable mental model of the ablation volume.[39] A contrasted T1-weighted MRI scan post-ablation characterises the ablation cavity, allowing removal of the laser probe and closure of the entry site. Patients typically recover overnight, greatly reducing hospital stay compared with an anterior temporal lobectomy.[40]

Seizure outcomes

LITT has been extensively studied as a treatment for mesial temporal lobe epilepsy, with 2-year seizure-freedom rates ranging from 53% to 62%.[41,42] A large prospective multicentre study in adults, in the USA, investigating the efficacy of LITT in mesial temporal lobe epilepsy has reported preliminary results (SLATE trial).[43] The outcomes for LITT are inferior to those for anterior temporal lobectomy, but comparable to those for selective amygdalohippocampectomy, which, similar to LITT, spares the temporal neocortex.[44] A 2025 meta-analysis of aggregate data from 42 studies including 1303 patients treated by LITT showed a seizure-freedom rate of 55·0% (measured at last follow-up, with 18 of the 25 studies included having >12 months of follow-up), a rate of major complications of 2·3% (defined as permanent visual field deficits, infections, or intracerebral haemorrhage), and a reoperations rate of 14·3%.[45] Existing literature on seizure outcomes after LITT with a minimum of 12 months' follow-up is summarised in the appendix (p 2). Additional studies with shorter follow up duration show concordant surgical efficacy and safety trends.[46–56]

Ideal candidates for LITT include those with unilateral hippocampal sclerosis and unilateral, ipsilateral epileptiform discharges. LITT can precede anterior temporal lobectomy in a staged approach, but subsequent open surgery might be riskier owing to scarring, adhesions, and altered anatomy. For hypothalamic hamartomas, LITT is considered the first-line therapy owing to the considerable morbidity associated with open and endoscopic approaches. 1-year seizure-freedom rates are between 68% and 93%, with particular benefit for gelastic seizures.[49,57–59] LITT for cavernous malformations can have 1-year seizure-freedom rates of over 80%.[60,61] LITT corpus callosotomy might reduce drop attacks, although studies are limited by small sample size and short follow-up.[46–48,62–64] In extratemporal lobe epilepsy, LITT shows promise for frontal lesional pathologies, with approximately 80% seizure freedom, but outcomes are poorer for people with non-lesional epilepsy, previous resections, or insular epilepsies.[65]

Neuropsychological outcomes

LITT might yield better short-term neuropsychological outcomes than open resections, and outcomes vary by target area and ablation extent, by sparing surrounding language and memory structures and pathways,[66] especially in mesial temporal lobe epilepsy. A meta-analysis of ten studies with 836 patients using reliable change indices showed postoperative declines in verbal memory in 24·2% of patients (95% CI 8·6–52), in visual memory in 25·2% of patients (8·3–55·8), and in naming in 13·4% of patients (6·6–25·4), regardless of laterality in patients undergoing LITT for temporal lobe epilepsy.[67] These declines were less frequent than those reported after open resection. Cognitive outcome data after LITT are scarce and variable, especially for extra-temporal lobe epilepsy.[67]

LITT might improve mood and quality of life, but residual cognitive deficits from ongoing seizures remain a concern for patients who do not reach seizure freedom. Overall, LITT offers a promising balance between seizure control and cognitive preservation. Currently, data on long-term neuropsychological outcomes after LITT are unavailable.[14,68,69]

Neuromodulation

Overview of surgical technique

When presurgical testing establishes multifocal or diffuse epilepsy, or the epileptogenic zone overlaps with eloquent cortex, neither resection nor ablation is an option. Such patients might be candidates for neuromodulation, including responsive neurostimulation, deep brain stimulation, and vagus nerve stimulation.[70] All involve implantation of stimulating electrodes connected to an internal pulse generator that supplies intermittent electrical stimulation to the target structure. Deep brain stimulation and vagus nerve stimulation are US Food and Drug Administration (FDA) and EU approved; responsive neurostimulation is only FDA approved. All three are approved for adults with focal drug-resistant epilepsy. Vagus nerve stimulation is also approved for children 4 years and older with focal or generalised epilepsies or Lennox-Gastaut syndrome. Off-label use in generalised and poorly localised epilepsies is expanding,[71] with ongoing clinical trials.[72,73]

See Online for appendix

Vagus nerve stimulation delivers stimulation via an electrode typically wrapped around the left vagus nerve,

	N	Study type (phase)	Age, years	Follow-up duration, years	Mean or median seizure reduction (IQR)	Patients with ≥50% reduction	Key points
Responsive neurostimulation							
Nair et al (2020; USA)[81]	162	Prospective open label (long-term follow-up)	≥18	9	75% (45–95)	73%	Follows previous trial patients (all trials are completed); adverse events: infection rate per neurostimulator 4·1%, non-seizure-related haemorrhage 2·7%, lead revisions 2·7%, subdural haemorrhage 2% (none had long-lasting neurological consequences); follow-up rate 89%
Deep brain stimulation							
Salanova et al (2021; USA)[82]	73	Prospective open label (long-term follow-up)	≥18	7	75%	74%	Follows previous trial patients; adverse events: memory impairment 30%, depression 37%, and suicidality in 10%; follow-up rate 62%
Peltola et al (2023; 13 countries)[83]	170	Observational registry (phase 4)	≥18	5	55%	53%	Adverse events: increased seizures 16%, memory impairment 15%, and depression 13%, with one SUDEP; 92·4% follow-up at 2 years and 53·2% at 5 years
Vagus nerve stimulation							
Kawai et al (2017; Japan)[84]	333	Large prospective registry (phase 4)	≥1	3	66%	59%	>90% follow-up rate

Open-label study or registry studies for neuromodulation therapies with more than 12 months' follow-up, where available. All studies had the same endpoints (mean or median seizure reduction and responder rate with patients with ≥50% reduction in seizures). SUDEP=sudden unexpected death in epilepsy.

Table: **Key studies examining seizure outcomes with neuromodulation techniques for drug-resistant epilepsies**

placed through an incision in the neck, connected to an internal pulse generator implanted in the chest.[74] Open-loop stimulation is preset at regular intervals with additional stimulation triggered manually by a magnet-swipe or by presumed ictal tachycardia detected by a closed-loop device.[75]

Responsive neurostimulation provides closed-loop stimulation in response to presumed epileptic discharges detected from continuous electrocorticography recorded by two four-contact depth or strip electrodes.[76] Used in multifocal or eloquent focal epilepsy, the electrode implantation targets the epileptogenic zones, and the internal pulse generator is situated in a craniectomy defect and contains electrocorticography data that can be downloaded wirelessly by the patient or provider to guide programming decisions.

Deep brain stimulation administers open-loop stimulation to symmetrical subcortical, typically thalamic, targets at regular preprogrammed intervals.[77] Although only implantation of the anterior nucleus of the thalamus is FDA approved, targeting other thalamic nuclei is becoming popular. Two four-contact or eight-contact depth electrodes are inserted and connected to a pulse generator implanted in the chest.

Other thalamic nuclei—centromedian, mediodorsal, and pulvinar—are being explored with both responsive neurostimulation and deep brain stimulation.[78] Additionally, a novel approach of epicranial-electrical focal cortex stimulation is under investigation, with early results showing a responder rate of 53% after 6 months of therapy.[79]

Seizure outcomes

Unlike with resection or ablation, seizure freedom is rare with neuromodulation. The therapeutic goal is to reach a response rate (responder-reduction in seizure frequency of ≥50%) that achieves an improvement in quality of life. With progressively titrated current delivery and personalised programming adjustments over time, all three neuromodulation devices show a temporal improvement in seizure control for the first few years, along with a poorly understood long-term neuromodulatory effect.[80] The trials for responsive neurostimulation and deep brain stimulation, including a 7–9-year prospective open-label study and vagus nerve stimulation data from retrospective or registry studies with a minimum 12 months' follow-up, are summarised in the table.

Responsive neurostimulation and deep brain stimulation trials showed 38–44% reductions in seizure frequency after a brief 3-month randomisation period. In the responsive neurostimulation open-label observational studies, the median seizure reduction was 53% at 2 years, 66% at 5 years, and 75% at 9 years.[81] With anterior thalamic deep brain stimulation, median seizure reduction was 56% at 2 years, 65% at 5 years, and 75% at 7 years.[82] For vagus nerve stimulation, median seizure reduction of 28–30% was noted after a 12–14 week randomisation period in trials examining both adults and teens.[74,85] A 2022 meta-analysis found that five observational studies of

vagus nerve stimulation in adults from multiple countries reported a pooled mean percentage decrease in seizure frequency of 35% at a mean 1·3 years,[70] and early data from a large prospective registry of vagus nerve stimulation outcomes reported 66% seizure reduction at 3 years; however, its observational design, potential selection bias, and follow up bias limit the strength of these conclusions. A reduction in seizure frequency of 50% or more, often used as the success measure for neuromodulation studies, is reached in 90·4% of patients who continue to seize after temporal lobe epilepsy resection, suggesting that even without complete resolution of seizures, resection offers optimal seizure control.[24]

Although several meta-analyses and systematic reviews have been published,[86–90] no prospective or rigorous observational head-to-head comparison of the effectiveness and safety of the three neuromodulation approaches is available.

Neuropsychological outcomes

There is a scarcity of robust data on the neuropsychological effects of neuromodulation techniques. Outcomes following deep brain stimulation are mixed; some studies report improved mood and quality of life, whereas others note potential cognitive side-effects, dependent on stimulation parameters and brain region.[91] Responsive neurostimulation has shown improvements in self-reported cognitive function in some patients in the long-term outcome study, probably due to reduced seizure activity.[81] The cognitive effects of vagus nerve stimulation are thought to be neutral, although improvements in mood can sometimes mediate improvements in cognitive function. Transient vagus nerve stimulation side-effects, such as hoarseness, can affect concentration and confidence. Overall, neuromodulation techniques have relatively favourable neuropsychological profiles compared with resective surgeries.

Integrating the concept of an epilepsy network in surgical practice

Conceptualising epilepsy as a disorder of atypical structural and functional brain networks has several implications. Starting with intracranial EEG evaluations, the network concept influences the choice of intracranial EEG technique and electrode placement. For example, if the hypothesised epileptogenic zone is a clearly visible malformation of cortical development abutting the motor cortex, a resection guided by intraoperative electrocorticography, with or without previous subdural electrodes, might suffice: the brain region of interest is limited, and residual epileptogenicity can be monitored intraoperatively while motor mapping ensures preservation of function. Conversely, if MRI does not detect a lesion, and the hypothesised epileptogenic zone is posterior perisylvian and insular with unclear lateralisation, a stereo-EEG allows safe exploration of such widely distributed networks. An international observational comparative effectiveness study found 1·66-times higher odds of seizure freedom with stereo-EEG than with subdural electrodes, and subdural electrodes had 2·24-times higher risk of complications than subdural electrodes.[92]

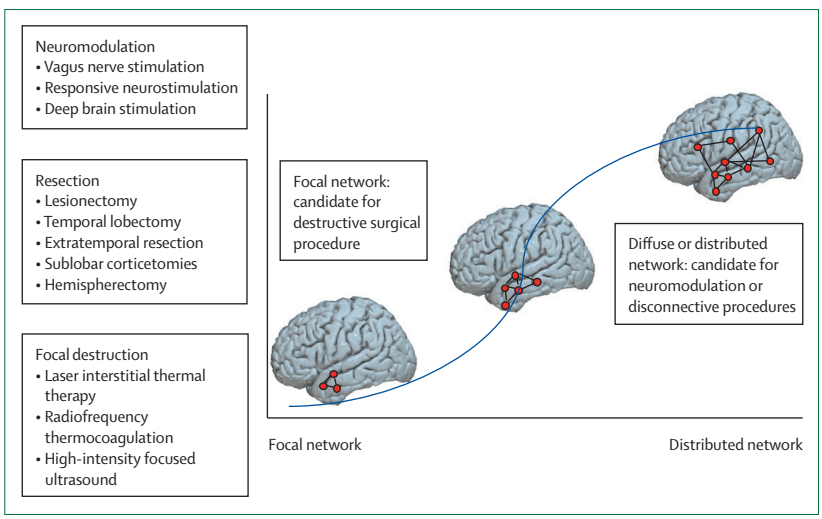

Figure 1: **A simplified approach to surgical decision making**
The illustration is based on literature correlating the extent of the epileptogenic network with surgical outcomes.[93–96] When multimodal testing shows a very focal epilepsy network, a small focal destructive procedure (eg, laser ablation) would have a 53–62% chance at 2-year follow-up of controlling seizures with minimal risk of postoperative memory decline (appendix pp 3–4). At the other extreme, neither resection nor ablation are expected to stop seizures if the epilepsy network is very distributed, making neuromodulation a more appropriate therapy. Understanding these advancements and incorporating measures of focality from multimodal imaging and EEG is essential to optimise patient selection and ensure timely referrals.

Once the hypothesised epileptogenic zone is delineated, the choice of resective or neuromodulatory therapies is similarly influenced by the epilepsy network identified (figure 1). Traditionally, epilepsy surgery focused on localising the epileptogenic zone and removing it, expecting complete seizure freedom. Current innovations are focused on addressing epilepsies with distributed networks where neither resection nor ablation is possible, and seizure palliation could still be worthwhile. A recent example of these rapidly shifting approaches is thalamic nuclei sampling with stereo-EEG,[97] to study their role in seizure generation and spread. Both qualitative visual ictal EEG review and analysis of quantitative corticothalamic evoked potentials show that thalamic nuclei other than the anterior nucleus could be involved early in seizure propagation.[98] Another new approach is using radiofrequency thermocoagulation during stereo-EEG evaluation to disrupt key network nodes, offering both therapeutic benefit and a preview of potential surgical success.[99] A meta-analysis studied 437 unique patients from 12 non-overlapping cohorts and analysed for seizure and complication outcomes after radio-frequency thermocoagulation: it documented a 1-year pooled seizure-free rate of 37·8% with a 1-year seizure responder rate of 69·7% (defined as ≥50% reduction in seizure frequency from baseline).[100] Patients with nodular heterotopia had higher rates of seizure freedom (57·1%)

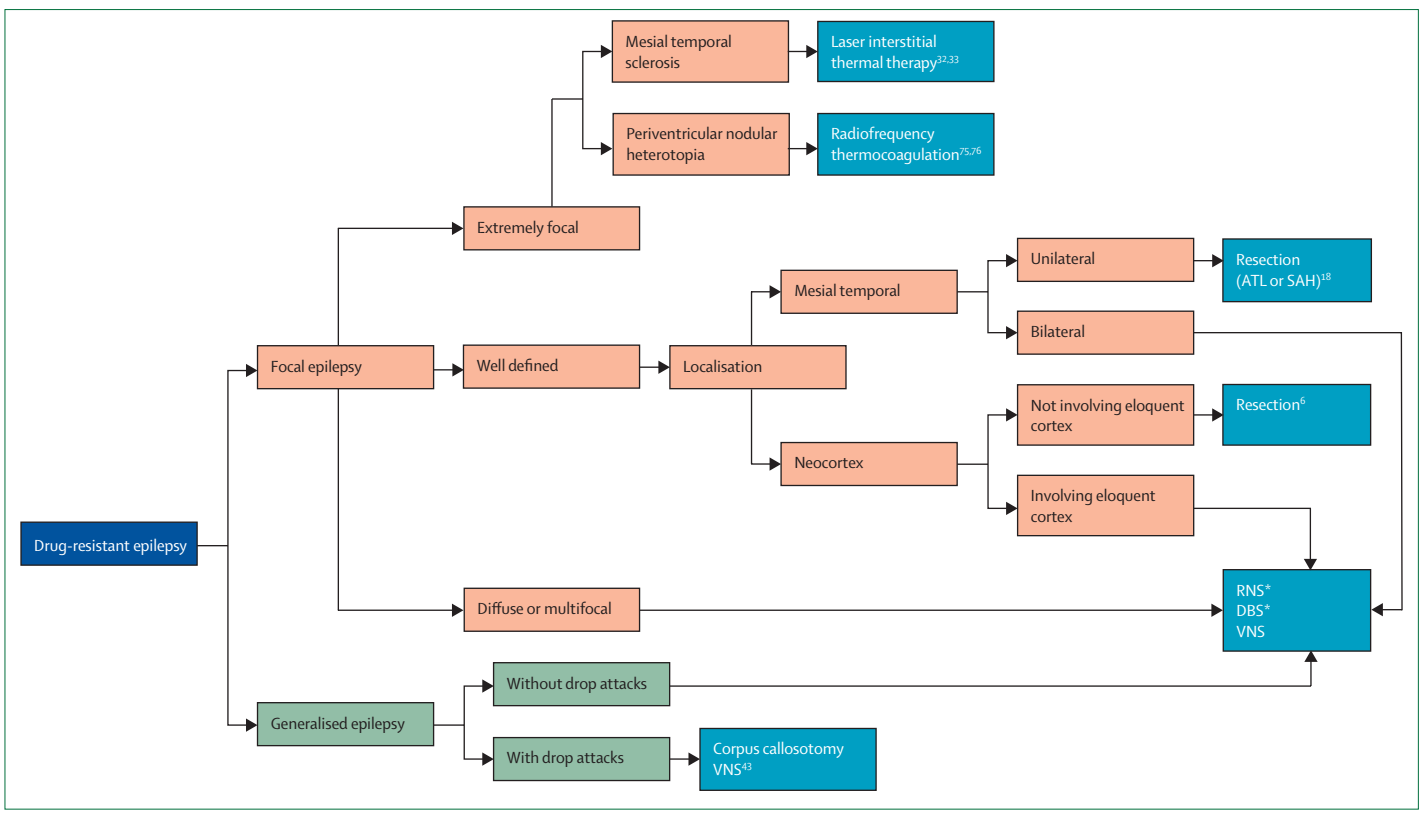

Figure 2: **Treatment algorithm**
The decision tree is synthesised from current societal guidelines, comprehensive literature review, and accepted clinical practice, offering structured guidance on the decision-making process for evaluating various surgical options across different clinical scenarios. Besides this outline, the decision-making process might need to factor in surgical risks based upon anatomical location, patient preferences, neurocognitive risks, and availability of resources. Orange represents focal epilepsy. Green represents generalised epilepsy. Light blue indicates available surgical techniques. ATL=anterior temporal lobectomy. DBS=deep brain stimulation. RNS=responsive neurostimulation. SAH=selective amygdalohippocampectomy. VNS=vagus nerve stimulation. *Under investigation for off-label use in generalised epilepsy.

than those with cortical dysplasia (12·0%) and MRI-negative conditions (24·6%). Similarly, patients with hippocampal sclerosis exhibited higher seizure-freedom rates (66·7%) than those with cortical dysplasia (12·0%).[100] The stability of this short-term response over longer follow-up needs to be further investigated.

Quantitative network analysis of brain imaging and EEG could aid in understanding epilepsy and guiding surgery. In network analysis, nodes are defined by different brain regions, and edges represent the connections between these regions.[101,102] Edges can be functional (eg, resting state functional MRI [rsfMRI], iEEG, and magnetoencephalography) or structural (eg, white matter connections estimated from diffusion tensor MRI [DTI]). Advanced computational methods can examine interactions and connectivity patterns among different brain regions that might otherwise be overlooked by traditional analyses.

Studies using network modelling in epilepsy surgery have focused on predicting outcomes from resection and neuromodulation. In destructive surgical procedures, widespread abnormalities in structural networks based on DTI and functional networks based on rsfMRI and stereo-EEG correlate with postoperative seizure recurrence, especially when abnormal regions were not resected.[103–108] Functional network measurements acquired during resting state magnetoencephalography or stereo-EEG have shown promise in predicting improvement in seizure frequency with responsive neurostimulation.[109,110] Overall, these studies show the potential of using structural and functional brain network measures to map patient-specific network perturbations and guide treatment decision making.

Multiple barriers hinder the integration of network-based analyses into clinical practice.[111] To bring network-based approaches to the bedside, large datasets to support validation are needed, along with user-friendly, preconfigured turn-key analytic tools that can be integrated into clinical workflows. Comparing network metrics to traditional clinical metrics will be key to showing their added value for clinical use. As treatment options for drug-resistant epilepsy grow, validated quantitative tools to guide clinical decision making will be crucial. For now, the practical translation of how these concepts can influence a surgical strategy is highlighted in figure 2.

Predicting outcomes

The ideal timing to initiate a surgical evaluation has been extensively debated, with recommendations by the International League Against Epilepsy (appendix p 5) emphasising the importance of early referral,[112] particularly in the paediatric and young adult population.[113] Most epilepsies are lesional in this population (eg, focal cortical dysplasia, epilepsy-associated tumours) and, therefore, patients are at higher risk of developing drug resistance, and timely elimination of seizures is crucial to optimise brain development and cognitive outcomes. Epilepsy surgery has a perioperative mortality rate of less than 1%, and a 2022 meta-analysis of prospective and retrospective cohort studies revealed that operated patients have a two-fold lower overall mortality risk (odds ratio 0·40, 95% CI 0·29–0·56; p<0·0001) and a three-fold lower sudden death in epilepsy risk (OR 0·32, 95% CI: 0·18–0·57; p <0·001) than non-operated patients, further supporting timely surgical evaluation.[114]

Determining surgical candidacy for any individual patient is complex. Clinicians integrate results from an ever-expanding set of imaging and electrophysiological studies, balancing epileptogenic zone localisation with neurocognitive sequelae of brain resection or ablation, explaining the low predictive accuracy of professional opinion.[115] Variable success rates across patient profiles and surgical techniques further complicate decision making.[116] Machine learning and outcome prediction tools are making surgical planning a more calculated and precise process. The first epilepsy surgery nomogram, a graphical calculation tool presented as a set of aligned scales representing complex mathematical relationships between variables, was introduced in 2015 with a low prediction accuracy (C-statistic 0·59).[21] The nomogram has since evolved to incorporate clinical, electrophysiological, imaging, neuropsychological, and genetic data to predict seizure outcomes (0·72), memory decline (0·81), and surgical complications with improved precision.[31,117–120] These tools, easily deployed as online risk calculators, enhance presurgical risk stratification. Similar calculators guide medication withdrawal after epilepsy surgery, with predictive accuracy (0·67).[121]

Nomograms rely on binary or simplified inputs, limiting their prediction potential by failing to capture nuanced data. Machine learning is addressing these limitations by automating data extraction from electronic medical records, enabling the use of large, longitudinal datasets.[122] Deep learning methods have shown great potential in epilepsy diagnostics. For example, standardised computer-based organised reporting of EEG-artificial intelligence is a convolutional neural network for the interpretation of scalp EEG, and has matched human-expert accuracy.[123] Machine learning models have effectively identified hippocampal sclerosis on MRI in people with temporal lobe epilepsy.[124,125] In parallel, several machine learning algorithms predict surgical outcomes leveraging DTI,[126] iEEG network measures, or PET imaging with moderate to high accuracy (area under the curve [AUC] 0·62–0·88).[127,128] The most accurate predictive machine learning algorithm to date analyses peri-ictal scalp EEG to predict temporal lobe epilepsy surgical outcomes with an AUC of 0·98 and out-of-group testing accuracy of more than 90%.[129] Pursuing more predictive models with routinely acquired non-invasive data will be key to expanding applicability to a wider patient population beyond select tertiary care referral centres in developed countries. Partnering with industry to integrate these models within health-care technologies (eg, EEG acquisition machines or MRI interpretation software) would facilitate integration in clinical workflows and improve clinician adoption. Figure 3 illustrates how presurgical diagnostic tools now influence decision making.

Integrating genetic and multi-omic analyses in epilepsy surgery

Rare genetic variants distinguished 206 people with temporal lobe epilepsy from 8718 healthy controls, but did not predict surgical success, in one observational case control study across North America, Europe, and Australia.[130] However, there are emerging data on specific genetic mutations and their influence on surgical practice. A recent systematic review and meta-analysis found postoperative seizure freedom was less likely with channel function and synaptic transmission gene mutations (12% seizure free), and better with mTOR pathway (58% seizure free), or other chromosomal abnormalities (neurofibromatosis type 1, fragile-X syndrome, or mitochondrial mutations [63% seizure free]).[131] Real-world data now link genetics to surgical decisions in paediatric drug-resistant epilepsy.[132] In a study of 16 patients with temporal lobe epilepsy (age 16–62 years) undergoing stereotactic laser amygdalo-hippocampotomy, RNA sequencing of peripheral leukocytes identified 24 differentially expressed genes distinguishing seizure-free from non-seizure-free outcomes. These multi-omic variations correlated with surgical outcomes and highlighted pathways in lipid metabolism, oligodendrocyte morphology, inflammatory response, and astrocyte development, supporting the emerging concept of neurosurgical genomics.[133] Genetic loci linked through independent pathway analysis with neuroinflammatory pathways and brain healing were similarly linked to late seizure outcomes.[134] Another study linked favourable seizure outcomes to expression of alternate alleles of three single nucleotide polymorphisms (rs10 276 036, rs11 975 994, and rs1 128 503) of *ABCB1,* a gene coding P-glycoprotein, which is involved in transport of antiseizure medications in drug-resistant epilepsy, and also to transport of dexamethasone, which is routinely used to reduce perioperative inflammation in the context of epilepsy surgery.[135] Median postoperative time-to-seizure was 63 months for patients with two alternate alleles,

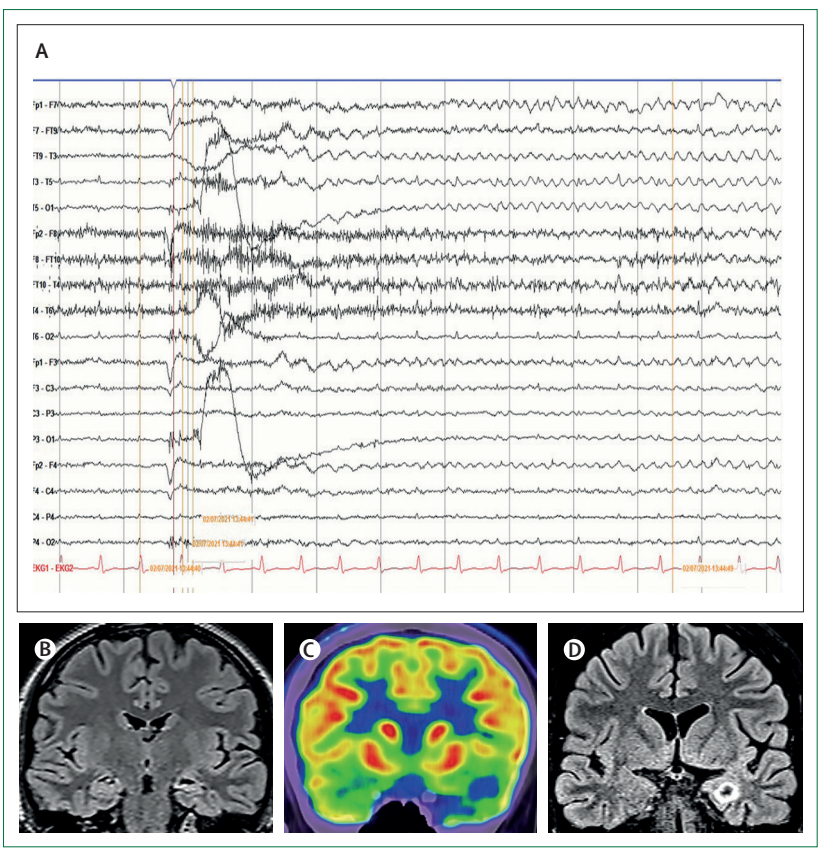

Figure 3: **Case study**
A woman aged 25 years, who was right-handed, with medically refractory epilepsy (three anti-seizure medications had no effect), with seizure onset at age 12 years, experiencing two to three focal seizures with impaired consciousness per month presented to the epilepsy clinic at one of the institutions where the authors are affiliated in June 2023. Seizures were described by the patient as "thick feeling", "stomach-dropping", and "hot flash", which were followed by staring, speech arrest, left-hand automatisms, and oral automatisms. (A) EEG captured seven electroclinical left temporal seizures with interictal discharges in the same distribution. MRI revealed left mesial temporal sclerosis (B) with concordant hypometabolism on [^{18}F]fluorodeoxyglucose-PET (C). Neuropsychological testing showed deficits in verbal and visual memory delayed recall, with better performance on visual memory acquisition than verbal acquisition and intact language functions: Rey Auditory Verbal learning test delayed recall raw score=4; Boston naming test raw score=53; preoperative F-A-S verbal fluency raw score=32; preoperative Wechsler Memory Scale-III verbal paired associates scaled score=1. Upon review of presurgical data by a board comprising of epileptologists, a neuropsychologist, a neurosurgeon, and a neuroradiologist at the institutional epilepsy surgery conference in the absence of the patient, the left mesial temporal structures were implicated and surgical options discussed (appendix pp 3–4). The consensus among the board of physicians was to offer a staged surgical approach, starting with minimally invasive MRI-guided LITT targeting the left hippocampus and amygdala to preserve memory. If LITT failed to control seizures, a standard anterior temporal lobectomy would be considered, offering higher seizure freedom but greater cognitive risk. The patient opted for a LITT procedure with post-procedure MRI (D) and she remains seizure free 8 months postoperatively. LITT=laser interstitial thermal therapy.

24–33 months with one alternate allele, and 10–11 months with no alternate alleles. A genetically driven variation in response to perioperative anti-inflammatory medications could translate to variation in the extent of surgical bed scarring and subsequent epileptogenicity. In 2023, a clustering technique was used to separate individuals into genetic clusters, and two molecular epilepsy cluster subtypes were identified through an independent analysis of blood-derived micro-RNA and mRNA expression levels, which were linked to poor seizure outcomes after temporal lobe epilepsy surgery.[136] Although no individual micro-RNAs were linked to outcomes, enrichment of a cancer drug-efflux resistance pathway involving hsa-miR-410 and hsa-miR-411 supported the involvement of efflux pumps. Targets identified in the two relevant endophenotypes were quantifiable in preoperative blood samples and were simultaneously represented in brain tissue, supporting the potential of blood-based biomarkers for temporal lobe epilepsy surgery outcomes.[136] Future efforts to better understand the molecular drivers of surgical outcomes could identify patient subgroups with targets for novel drugs and therapies and improve the rates of postoperative seizure freedom.

Disparities in access and use

Epilepsy surgery in low-income and middle-income countries

Low-income and middle-income countries (LMICs), as defined by the World Bank 2024 fiscal year, face substantial treatment gaps in medical and surgical epilepsy care, especially with presurgical testing.[137–139] Only 20% of LMICs report conducting epilepsy surgery, primarily due to geographical barriers, poor affordability, insufficient resources, and a low physician-to-patient ratio.[140] Patient and caregiver factors, such as low awareness, misinformation, fear of surgery, and myths about epilepsy, further exacerbate these challenges. There is a shortage of trained neurosurgeons, neuroradiologists, and epileptologists, and equipment such as video EEG, structural brain MRI, and advanced imaging facilities are often unavailable or inaccessible. These barriers contribute to long waiting times for people with drug-resistant epilepsy seeking surgical therapy.[141–143] Most LMICs have few or only one centre performing epilepsy surgery, requiring people to travel long distances for evaluation and surgical visits. Neurologists and neurosurgeons in these regions often train in other countries and later establish epilepsy programmes in their home countries, with initial mentorship via virtual conferences. Early stage programmes can benefit from focusing on clear and well localised epilepsies, including concordant cases with lesions, such as hippocampal sclerosis, developmental neuroepithelial tumours, or clear structural epileptic lesions. As expertise and confidence grow, more challenging cases can be tackled, including type 1 focal cortical dysplasia or epilepsies with normal MRI findings. Stereo-EEG is available only in a handful of centres in LMICs, restricting the ability to evaluate surgical options in complex epilepsies.[144,145]

Disparities in epilepsy surgery practices

Disparities in epilepsy surgery extend beyond LMICs to include high-income countries where socioeconomic status, racial and ethnic differences, payer status, and access to care contribute to delayed referrals and underuse of surgery.[146–149] Cultural barriers,

Panel 2: **Research gaps and priorities in the field of epilepsy surgery**

Treatment options and decision making

Knowledge gaps

- Shortage of rigorous comparative effectiveness analyses between resection, ablation, vagus nerve stimulation, responsive neurostimulation, and deep brain stimulation
- Poor understanding of the optimal surgical techniques for emerging non-resective approaches (eg, ideal probe targeting with laser ablation, or data-driven stimulation parameter setting for responsive neurostimulation or deep brain stimulation)
- Identifying the best surgical approach for any given patient is still largely dependent on physician interpretation of presurgical testing

Proposed research priorities

- Expand scope of research into mechanisms of seizure-recurrence to include molecular and genetic factors
- Randomised clinical trials or carefully designed observational studies to compare seizure outcomes and risks of available interventions
- Large surgical registries tracking technical details of surgical procedures, patient characteristics, presurgical data, and outcomes, as is done in other disciplines (eg, registry of the Society of Thoracic Surgery)[154] to enable surgical practice outcomes and clinical practice research
- Improve adoption of implementation science and data-driven clinical decision support research

Predicting outcomes

Knowledge gaps

- Minimal prospective validation of predictive tools
- Advanced predictive models either use invasive data recording techniques (intracranial EEG) or complex multimodal imaging, restricting integration in clinical practice
- Molecular predictors are still in the early stage

Proposed research priorities

- Prospective validation of available predictive tools, focusing on reproducibility in external cohorts and resource-poor settings
- Explore predictive benefits of more widely available non-invasive data recording techniques (eg, scalp EEG).
- Partner with industry (eg, neuroimaging equipment manufacturers) to integrate imaging-based predictive models into user-ready software that can be built into imaging machines
- Increase research focus on the role of cellular mechanisms in defining outcomes (eg, multi-omic analyses)

Disparities in access and use

Knowledge gaps

- Infrastructural inequities persist
- Cost and training barriers in low-income and middle-income countries for new technologies
- Effects of these technological advances on access are unclear

Proposed research priorities

- Develop and test strategies that bridge diagnostic and treatment gaps, rather than continue to do research that documents inequities
- Use of portable devices such as ultra-low field MRI in presurgical planning[6]
- Study the role of digital tools (eg, telemedicine, artificial intelligence-enabled EEG interpretation) in expanding surgical care access

misconceptions, and perceiving surgery as a last resort hinder adoption worldwide.[150]

The availability, expertise, and payer coverage of surgical techniques vary substantially across regions, complicating efforts to standardise care. In LMICs, anterior temporal lobectomies are predominantly done in straightforward cases, and surgeries for patients with more challenging non-lesional epilepsies and neuromodulation procedures are done less frequently owing to infrastructural constraints.[151,152] This treatment gap results in less favourable post-surgical outcomes in lower socioeconomic groups than in higher groups.

Overcoming these barriers requires a multifaceted approach. Providers can benefit from educational outreach, incentives for timely referrals, and structured referral pathways that involve multi-disciplinary teams promoting collaboration between LMICs and high-income countries.[112] Remote surgical training through real-time streaming and virtual simulators might expand expertise and increase the number of epilepsy surgery centres in LMICs. For patients, confidence might be strengthened through self-management programmes, support groups, and increasing racial diversity among care providers. Access can be improved through telemedicine, mobile clinics, and transportation support, and the implementation of national policies to make these services reimbursable.

Emerging technologies, such as machine learning for surgical candidate identification from electronic medical records, could facilitate timely referrals and reduce implicit biases.[122] Artificial intelligence models are being developed to automate EEG and imaging interpretation, which could improve diagnostic outreach in underserved areas.[123–125] Minimally invasive epilepsy surgery might make surgery more acceptable to patients, though its cost-effectiveness and insurance coverage remain uncertain without robust evidence of efficacy.[153]

Search strategy and selection criteria

We conducted a systematic search of literature on advances in epilepsy surgery and outcomes. We searched MEDLINE and PubMed used terms such as "epilepsy surgery," "drug-resistant epilepsy," "resective surgery," "laser interstitial thermal therapy", "neuromodulation", and "minimally invasive techniques" with Boolean operators (AND, OR). The search included peer-reviewed English articles published from Jan 1, 2019 to March 31, 2025, excluding non-English articles. Additional articles were identified by manually reviewing references of selected studies and key reviews.

Conclusions and future directions

Epilepsy surgery is evolving towards less invasive, personalised, network-based therapies. Resective surgery remains the most definitive curative approach, with ongoing efforts to improve outcomes via early diagnosis, equitable access, better localisation, and molecular biomarkers. Refining LITT approaches can offer potentially safer alternatives to some patients. Comparing outcome drivers of neuromodulation devices is paramount, considering their variable costs and risks. Future research (panel 2) and innovation will require multidisciplinary teams with skillsets in molecular biology, computational neurosciences, biomedical engineering, advanced imaging expertise, and clinical and surgical epileptology.

Contributors

All authors contributed to the conceptualisation, data curation, writing of the original draft, and review and editing of the final Review.

Declaration of interests

KD participates on advisory boards for NeuroPace, Rapport Therapeutics, Neurelis, Biocodex, UCB Pharma, UNEEG medical, Spark Therapeutics, Azurity Therapeutics, and Encoded Therapeutics. DE has received consulting fees from NeuroPace, a pending US patent application (number 18/676,682) regarding adaptive neuromodulation in epilepsy developed academically through Vanderbilt University with no monetisation currently planned, and is on the data safety monitoring boards for NeuroPace, UT Houston, and the University of California, San Francisco. CW receives grants from Dixi Medical, receives consulting fees from BrainLab, Neuropace, Medtronic, NeuroOne, and Renishaw, and is on the board of directors for the American Society for Stereotactic and Functional Neurosurgery. LJ is on the board of the American Epilepsy Society. All other authors declare no competing interests.

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